Race, explain and transfer

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Evaluate a model against measured performance and defend design decisions.

SUCCESS CRITERIA - I can...

1. Report measured distance, time and average speed.
2. Compare actual speed with a stated prediction.
3. Use evidence to explain three physical design choices.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A 60-tooth driver turns a 480-tooth gear. Give ideal output/input torque and speed ratios and state the loss assumption.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: average speed = distance/time
relative deviation = |measured - predicted|/|predicted| × 100%. Prediction must be nonzero. Record real race conditions; this compares with a prediction, not symmetric percent difference.

Worked reasoning: A car covers 3 m in 2 s: average speed 1.5 m/s. A complete explanation also uses its circuit, gearing, friction and evidence from repeated runs; one race time does not isolate a cause.

Your turn: Draw an energy-flow diagram from battery to motion and other transfers; annotate it with one actual result, clearly labeled as measured.

Marzano check - circle: ① ② ③

03 Represent and interpret

Draw or annotate your final car. Label three choices involving wheels, gears, alignment, wiring or mass, then connect each to a physics idea.

Marzano check - circle: ① ② ③

04 Solve and justify

A supplied car travels 6.0 m in 8.0 s; predicted speed is .90 m/s. Calculate measured speed, relative deviation and a plausible source of mismatch.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Analyze the reasoning

Use your actual race record to defend one successful design choice and identify one revision for a future build. Cite data or a specific observation for each.

Marzano check - circle: 1 2 3

06 Choose a method

A motor is 64% efficient and the following generator is 72% efficient. Find the combined useful fraction in sequence.

Marzano check - circle: 1 2 3

07 Exit check

Transfer: a similar car uses twice the wheel diameter with the same loaded wheel rpm. Predict ideal speed change, then explain why testing is still necessary.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Supplied input is 6 V×1 A for 4 s. A 0.5 kg car reaches 2 m/s on a level track from rest. Find input energy and final translational kinetic energy. Explain why their ratio alone is not total drivetrain efficiency.

Marzano check - circle: 1 2 3